Project Proposal

California Electricity Demand and Weather Analytics
Hourly CAISO grid load joined with population-weighted weather, 2015-2026

1. Team Members

Full Name	ID	Phase 1 Role
Hisham Wael Hosny	692300279	Dataset description & data dictionary
Zeyad Mohamed	692300066	Docker setup (PostgreSQL, pgAdmin) & GitHub repo & submission video
Mohamed Mostafa	692300110	Architecture diagram
Basel Tamer Fathy	692300055	Database table design (star schema)
Fatma Shrief	692300201	Dataset selection & proposal lead, Analytical questions (Q1-Q3)

2. Dataset and Source Description

Name	CAISO Hourly Demand + Population-Weighted Weather
Link	https://www.kaggle.com/datasets/arimancy/caiso-hourly-demand-weather
Publisher / License	Arimancy LLC / CC BY 4.0
Size	About 98,968 rows x 35 columns (hourly records)
Period	July 2015 to October 2026, no missing hours; imputed or preliminary values are flagged
Format	Parquet and CSV
Demand source	EIA-930 hourly grid data, cleaned by Catalyst Cooperative (PUDL), plus a short recent tail from the EIA API. CAISO covers roughly 80% of California load.
Weather source	ERA5 reanalysis (via Open-Meteo) at 7 cities: Los Angeles, Riverside, San Diego, San Francisco, San Jose, Sacramento, Fresno, combined using Census population weights.

Columns by data type

Type	Columns
Timestamp	datetime_utc, datetime_local
Float	demand_mwh, demand_reported_mwh, demand_imputed_pudl_mwh, temp_c_* (7 cities), temp_pop_wt_c, temp_pop_wt_f, shortwave_radiation_pop_wt_wm2, cloud_cover_pop_wt_pct, hdh_65, cdh_65, hdd_65_daily, cdd_65_daily
Integer	hour, day_of_week, month
Boolean	demand_is_imputed, temp_is_missing, temp_is_imputed, is_weekend, is_holiday_statutory, is_holiday_observed, is_holiday_ca_statutory, is_holiday_ca_observed, is_dst_transition_day, weather_is_preliminary
String	demand_source, demand_imputed_pudl_mwh_imputation_code

3. Analytical Questions

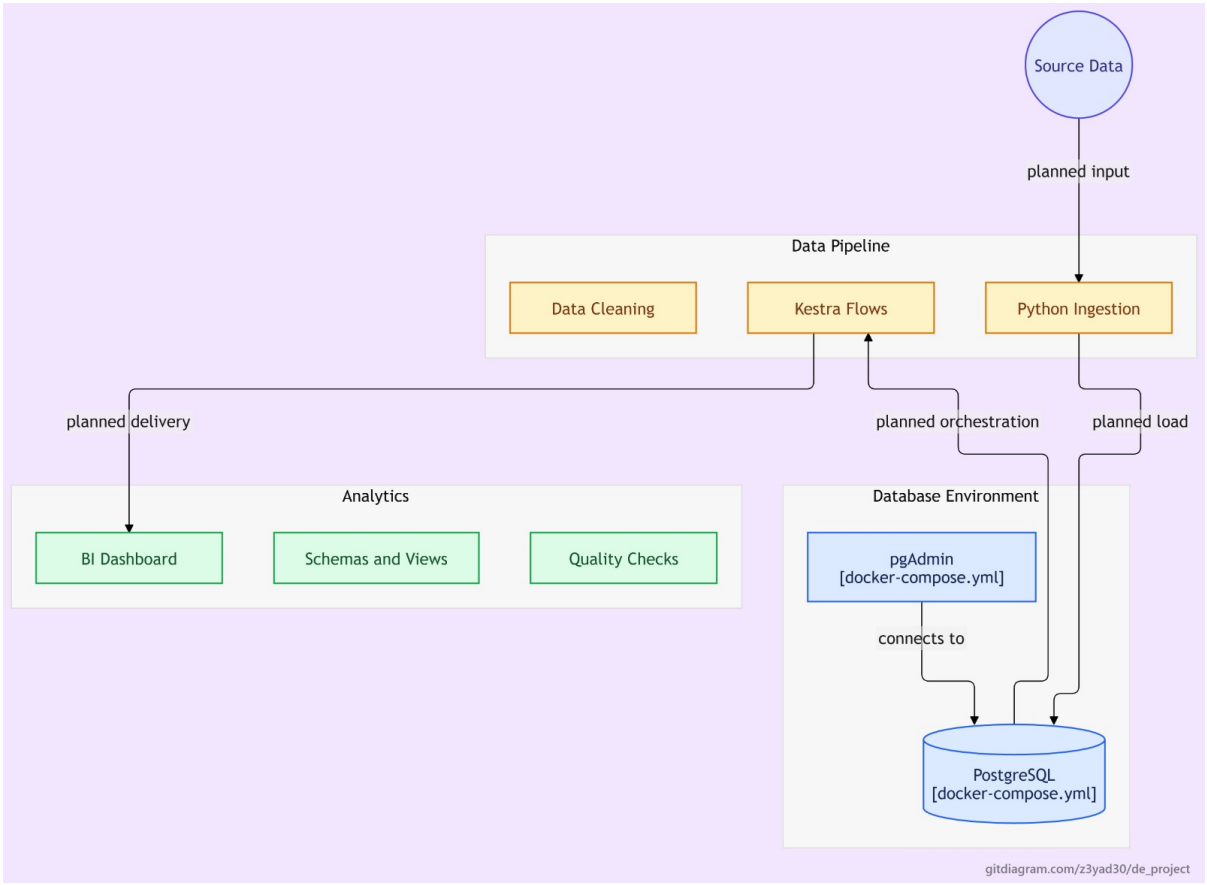
Q1. **Temperature effect:** How does temperature affect hourly electricity demand? At what temperature does demand start to go up fast?

Q2. **Solar and the daily load shape:** How has the midday dip in demand changed from 2015 to 2026 in each season? How do sunshine and cloud cover relate to it?

Q3. **Calendar and cities:** How do the hour of the day, weekends, and holidays affect peak demand? Which of the 7 cities' temperatures explains demand best?

4. Proposed Architecture

[Source] -> [Python Ingestion] -> [PostgreSQL Database] -> [Kestra Orchestrator] -> [Dashboard]



Everything runs locally with Docker Compose (PostgreSQL, pgAdmin, Kestra containers).

5. Environment Setup (Docker) and Repository

We use a **docker-compose.yml** file to start PostgreSQL and pgAdmin, and later Kestra. PostgreSQL keeps its data in a persistent volume, so nothing is lost when the containers restart. Passwords and settings stay in a local **.env** file, and only **.env.example** is pushed to GitHub. Each of us runs **docker compose up -d** on our own laptop.

Repository folder	Content
docs/	Proposal, architecture diagram, data dictionary
scripts/	Extraction, cleaning, and ingestion code (Python)
sql/	Table schemas (DDL), quality checks, and views
flows/	Kestra YAML orchestration flows
docker-compose.yml, .env.example, README.md	Environment setup and run instructions

6. Expected Database Tables and Columns

The raw CSV is first loaded into a staging table, then Kestra-orchestrated SQL splits it into a star schema:

Table	Type	Columns
raw_caiso_hourly	Staging	All 35 original CSV columns, loaded as-is by the Python ingestion script (untouched raw copy).
dim_datetime	Dimension	datetime_id (PK), datetime_utc, datetime_local, year, month, day_of_week, hour, season, is_weekend, is_dst_transition_day
dim_holiday	Dimension	datetime_id (FK), is_holiday_statutory, is_holiday_observed, is_holiday_ca_statutory, is_holiday_ca_observed
dim_city	Dimension	city_id (PK), city_name, population_weight
fact_demand	Fact	datetime_id (FK), demand_mwh, demand_reported_mwh, demand_imputed_pudl_mwh, imputation_code, demand_is_imputed, demand_source
fact_city_temperature	Fact	datetime_id (FK), city_id (FK), temp_c (7 city columns reshaped from wide to long)
fact_weather_weighted	Fact	datetime_id (FK), temp_pop_wt_c, temp_pop_wt_f, shortwave_radiation_pop_wt_wm2, cloud_cover_pop_wt_pct, hdh_65, cdh_65, hdd_65_daily, cdd_65_daily, temp_is_missing, temp_is_imputed, weather_is_preliminary